

MAT- 2137(ENT)

IA 4

1. Consider digraphs $29x + y$, in 29 letter alphabet (with “blank”=26, “full stop”=27 and “comma”=28) and encrypt your full name as in the Institute registration, using Affine transformation $C=AP+B$, by taking $A=\text{Roll number} + 11$ and $B=3x(\text{Roll number}+22)$, except for roll numbers 18 and 47. Also verify by decrypting back.
(For Roll Number 18, it is A= 37 and for roll number 47, it is A=86)
 2. Encrypt your name, by the method of encryption of digraph vectors over 27-letter alphabet (with blank=26) using a suitable invertible matrix (mod 27). Also verify your answer.
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